Department of Software Engineering
Faculty of Science & Information Technology

# UniExpert

## AI-Powered Expert Discovery and Academic Collaboration Platform

Capstone Project Proposal

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**Course:** SE 331 — Software Engineering Design Capstone Project

**Semester:** Fall 2026 · **Duration:** October – December 2026

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## Executive Summary

Undergraduate and graduate students undertaking capstone projects, theses, and innovative research consistently face a systemic operational bottleneck: identifying and initiating collaboration with domain-specific experts, interdisciplinary researchers, and experienced industry mentors. Traditional institutional directories, keyword-based search tools, and siloed academic websites fail to bridge this divide. Students often lack formal knowledge of academic taxonomic titles (e.g., knowing whether a problem requires a computer vision specialist, a biostatistician, or a bioinformatics researcher) and have limited professional networks, while experts are inundated with vague, poorly matched solicitations.

To address this challenge, this capstone proposal introduces **UniExpert**, an intelligent, AI-powered expert discovery and academic collaboration platform. UniExpert combines natural language requirement parsing, dense semantic embeddings, multi-criteria hybrid ranking, and an automated research team builder to connect student research problems with verified academic and industrial specialists across all major domains. Rather than searching solely by fixed keywords, students input unstructured, natural language problem descriptions. UniExpert extracts primary and secondary domains, technical skill prerequisites, and research scopes, subsequently mapping these requirements to a curated and verified expert corpus. Recommendations are ranked using a hybrid scoring model integrating domain relevance, skill overlap, research interest similarity, verified credentials, availability, and collaboration preferences, complete with explainable natural language rationale.

The engineering and research scope of this Fall 2026 capstone project (October – December 2026, 12 weeks) encompasses: (1) design and implementation of a scalable full-stack web application; (2) construction of a hybrid recommendation pipeline; (3) an empirical comparative study evaluating the hybrid engine against classical BM25 keyword matching and dense vector retrieval; and (4) structured verification and safety workflows tailored to academic integrity.

**Project Quick Reference Card**

| | |
|---|---|
| **Project Name:** | **UniExpert** (AI-Powered Academic Collaboration Platform) |
| **Target Users:** | University Students, Professors, Researchers, Industry Mentors, Admins |
| **Core Innovation:** | Problem-driven Natural Language Querying, Hybrid Multi-Criteria Matching, Multi-Tier Evidence Verification, and Explainable Team Assembly |
| **Primary Research Question:** | *Does hybrid AI-based expert matching provide statistically significant higher relevance and user conversion than keyword retrieval and semantic similarity alone?* |
| **Evaluation Metrics:** | Precision@5, Recall@5, NDCG@5, Skill Extraction F1, Latency, Conversion Rate |
| **Primary Deliverables:** | Full-stack platform, reproducible AI pipeline, benchmark dataset, research paper. |

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# 1 Project Overview and Problem Statement

## 1.1 Background and Academic Context

In modern higher education, academic projects—such as final-year capstones, honours theses, and independent research initiatives—are increasingly interdisciplinary. A biomedical device requires expertise in both embedded systems and physiological sensors; an agricultural prediction system combines machine learning with poultry pathology; and a financial analysis tool requires both software architecture and quantitative econometric modeling.

Despite this interdisciplinary reality, university mentoring structures remain overwhelmingly siloed within distinct academic faculties. Students are typically restricted to consulting the immediate faculty members of their home departments. When an academic project ventures beyond departmental boundaries, students face severe friction in discovering, vetting, and contacting appropriate experts.

## 1.2 The Core Problem Statement

The core problem addressed by **UniExpert** is formulated as follows:

> *University students frequently struggle to identify, evaluate, and establish collaboration with domain-specific experts, researchers, and professional mentors whose specialized knowledge is essential for the completion and academic rigor of their research, thesis, and capstone projects.*

This systemic failure stems from four distinct structural deficiencies:

**(1) The Vocabulary Mismatch Problem:** Students understand their project goals in plain language but rarely know the precise academic nomenclature, taxonomies, or specialized credentials required (e.g., searching for "poultry disease AI" instead of "computational epidemiology and computer vision for livestock welfare"). Keyword-based systems fail under this vocabulary divergence.

**(2) Fragmentation and Cold Outreach Overhead:** Expert information is scattered across disparate sources: institutional staff directories, personal websites, Google Scholar profiles, LinkedIn, and GitHub repositories. Students resort to speculative cold emails, which frequently lack context, leading to high rejection and ignore rates.

**(3) Absence of Verification and Credential Distrust:** Self-reported claims on generic professional networks cannot differentiate between actual peer-reviewed research publications, accredited institutional degrees, and superficial familiarity, exposing students to unverified advice.

**(4) Multidisciplinary Team Bottlenecks:** Multi-faceted projects require cohesive teams of complementary specialists rather than a single advisor. Students lack automated tools to detect team skill gaps and assemble multidisciplinary advisory panels.

## 1.3 The Proposed Solution: UniExpert

**UniExpert** resolves these challenges by introducing an end-to-end, AI-powered expert discovery and academic collaboration platform. The system operates on a *problem-first paradigm*: students articulate their project goals using unstructured natural language, and the underlying AI architecture parses, decomposes, and contextualizes these requirements. It matches the project against a database of verified experts from academia, research institutions, and industry, scoring candidates based on multi-dimensional compatibility and generating transparent, explainable recommendations.

### 1.4 Scope Boundaries and Non-Goals

A critical engineering principle for this capstone is strict scope discipline. UniExpert is deliberately designed with specific boundaries:

- **In-Scope:** Problem parsing, semantic search, hybrid multi-criteria ranking, explainable recommendation justifications, multi-tier credential verification, project postings, structured collaboration requests, and workspace communication channels.
- **Explicit Non-Goals (Out of Scope for MVP):** UniExpert will **not** attempt to be a general open social network, a freelance gig-work financial marketplace, a real-time multi-party video conferencing engine (Zoom clone), or a full-fledged laboratory management suite. Its sole focus remains on *accurate, trust-verified expert discovery and collaboration inception.*

## 2 User Roles and Stakeholder Architecture

The UniExpert platform defines four distinct user personas, each equipped with dedicated functional capabilities and role-based access control (RBAC) privileges, as illustrated in Table 1.

## 3 Functional Feature Specifications

The platform incorporates twenty-five distinct features categorized into Core Platform Capabilities, Advanced AI/ML Systems, Community Discovery, and Trust/Security Workflows.

### 3.1 Core Features

#### 3.1.1 Feature 1: AI-Powered Problem Analysis

Students input an unformatted natural language description of their intended research. An integrated Natural Language Processing (NLP) pipeline analyzes the submission to extract:

- **Main Problem Statement:** Synthesized core research inquiry.
- **Primary and Secondary Domains:** Taxonomic classification (e.g., Primary: Artificial Intelligence; Secondary: Poultry Science).
- **Required Technical Skills:** Specific tools, libraries, or methods (e.g., PyTorch, YOLOv8, Spectrometry).
- **Required Research Experience:** Academic level (e.g., peer-reviewed publication track record, clinical trial experience).
- **Recommended Collaboration Type:** e.g., Capstone Co-Advising, Methodology Review, or Technical Code Consultation.

**Case Example**

**Student Query Input:** *"I want to develop a machine learning system to predict poultry diseases, but I need someone experienced in machine learning and poultry science."*

**System NLP Output Analysis:**

- **Primary Domain:** Computer Science → Machine Learning → Computer Vision / Predictive Modeling
- **Secondary Domain:** Agricultural Sciences → Animal Science → Poultry Pathology
- **Extracted Skills:** Supervised Classification, Image Annotation, Veterinary Diagnostics, Sensor Telemetry

Table 1: UniExpert Role-Based Access Control and Persona Matrix

| Role | Target Persona | Core Capabilities & System Responsibilities | Auth Level |
|---|---|---|---|
| **Student** | Undergraduate & Graduate Researchers, Capstone Teams | • Create and maintain academic profile.<br>• Submit natural language problem statements.<br>• Execute semantic searches; receive hybrid recommendations.<br>• Construct multi-member research teams; detect skill gaps.<br>• Transmit structured collaboration invitations.<br>• Access project workspace and rate completed mentorships. | Standard (Tier 1) |
| **Expert** | Professors, Postdocs, Industry Specialists, Alumni Researchers | • Maintain verified credentials, publications, and portfolio links.<br>• Specify availability schedules and collaboration types.<br>• Receive, review, accept, or decline collaboration requests.<br>• Participate in workspace discussions and track milestones.<br>• Acquire peer reviews and institutional verification badges. | Verified (Tier 2) |
| **Institution Admin** | University Deans, Department Heads, Research Offices | • Manage institutional affiliation rosters and departmental tags.<br>• Verify institutional academic email domains and staff IDs.<br>• Monitor intra- and inter-institutional collaboration metrics.<br>• Review reported misconduct within institutional jurisdiction. | Delegated (Tier 3) |
| **Platform Admin** | Capstone Engineering Leads, Platform Operators | • Supervise global user accounts and audit system integrity.<br>• Curate taxonomy trees, technical skills, and domain registries.<br>• Review escalated fraud, impersonation, or academic dishonesty.<br>• Monitor ML model latency, recommendation drift, and system logs. | Superadmin (Tier 4) |

- **Suggested Collaboration:** Interdisciplinary Thesis Mentorship / Co-Supervisory Panel

#### 3.1.2 Feature 2: Intelligent Multi-Criteria Expert Matching

The platform produces an ordered ranking of candidate experts whose profiles, verified publications, skill matrices, and availability align with the extracted requirements. Every candidate is ranked via our hybrid scoring function (detailed in Section 4) and paired with an explainable justification.

#### 3.1.3 Feature 3: Semantic Natural Language Expert Search

In addition to project-based matching, students can perform ad-hoc semantic queries. The system utilizes dense vector embeddings to map natural language queries (e.g., *"Someone who can help me publish an AI research paper"*, *"A person experienced in solar energy optimization"*) into a continuous embedding space, matching profiles with semantic equivalence even in the presence of divergent terminology.

#### 3.1.4 Feature 4: Multidisciplinary Expert Discovery

When a research proposal spans non-overlapping technical domains, the system executes an automated problem decomposition. For a project combining *Computer Vision*, *Veterinary Medicine*, and *Cloud Deployment*, the engine automatically segments requirements into distinct specialized roles and suggests either a singular polymath researcher or a distributed advisory cohort.

#### 3.1.5 Feature 5: Comprehensive Expert Profile and Portfolio

Each expert profile hosts a structured, multi-dimensional academic dossier containing: professional headline, verified institutional affiliations, educational history, domain competencies, publication repository links (Google Scholar, ORCID, DBLP), active projects, verified certifications, availability windows, and past student feedback. Crucially, the UI distinctly segregates **independently verified records** from **self-reported claims**.

#### 3.1.6 Feature 6: Multi-Tier Expert Verification System

To guarantee institutional trust and combat academic fraud, UniExpert establishes a progressive four-tier verification framework:

1. **Tier 1: Institutional Email Verification:** Automated verification of educational domain emails (e.g., `.edu`, `.ac.uk`).
2. **Tier 2: Institutional Affiliation Audit:** Manual or delegated institutional admin confirmation of faculty status or graduate enrollment.
3. **Tier 3: Academic Publication Binding:** Cryptographic or OAuth linkage to verified ORCID / Google Scholar profiles with author cross-validation.
4. **Tier 4: Manual Document Validation:** Secure administrative inspection of official degree transcripts or professional certifications.

#### 3.1.7 Feature 7: Research and Project Posting Board

Students can publish detailed project opportunities with granular visibility controls (Public, Campus-Only, or Invitation-Only). Postings define research objectives, technical prerequisites, expected duration, deliverable milestones, and budget allocations (if applicable).

### 3.1.8 Feature 8: Collaboration Request Lifecycle Management

Students initiate contact through formal, structured collaboration requests. To eliminate low-effort cold emails, requests mandate a project abstract, expected deliverables, proposed time commitments, and meeting schedules. The request follows a deterministic Finite State Machine (FSM) depicted in Figure 1.

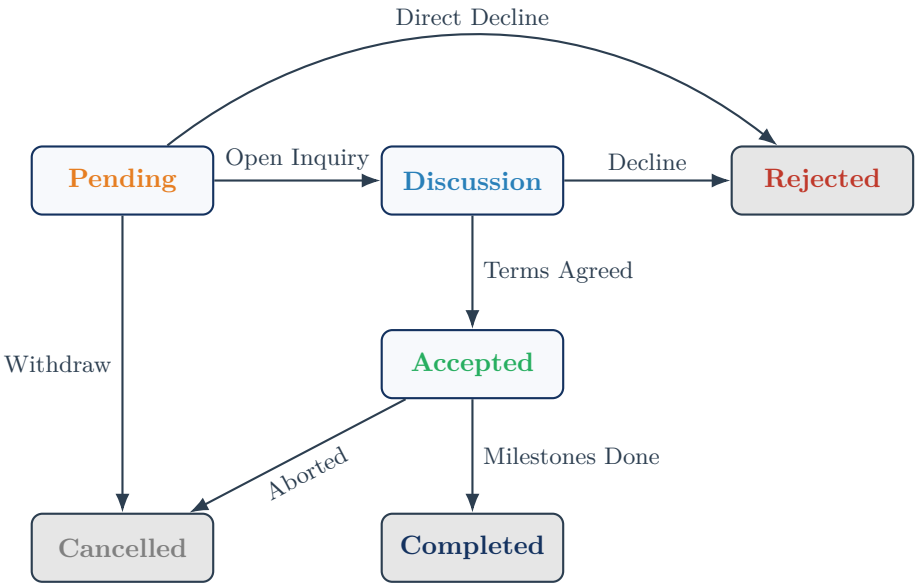



Figure 1: Finite State Machine for Collaboration Request Lifecycle

### 3.1.9 Feature 9: Research Team Builder

Allows students to configure required multidisciplinary roles for complex capstone projects. The AI engine evaluates prospective candidates simultaneously, optimizing for collective skill coverage, schedule compatibility, and complementary expertise while proactively alerting students to vacant competencies.

### 3.1.10 Feature 10: AI Research Requirement Assistant

An interactive conversational assistant designed to help novice students refine ambiguous research inquiries prior to expert outreach. The assistant identifies missing methodologies, breaks the project into structured milestones, and drafts professional, highly compelling collaboration proposals.

### 3.1.11 Feature 11: Side-by-Side Expert Comparison Engine

Provides students with an objective comparative dashboard evaluating candidate experts across orthogonal dimensions: skill overlap percentage, publication count in the specific subfield, verified status, availability score, and past student rating distributions.

### 3.1.12 Feature 12: Integrated Workspace and Discussion Channels

Upon request acceptance, the platform automatically provisions a secure collaboration workspace featuring thread-based asynchronous messaging, document and proposal attachment sharing, milestone tracking, and calendar meeting coordination.

# 4 Advanced AI/ML Recommendation Architecture

To fulfill the rigorous software engineering and computer science research requirements of a capstone, UniExpert introduces a multi-stage hybrid retrieval and ranking pipeline.

## 4.1 The Hybrid Matching Model Formulation

Given a student research problem profile $u$ and a candidate expert profile $e \in \mathcal{E}$, the ranking engine computes a composite compatibility score $\mathcal{S}(u, e) \in [0, 1]$ via a multi-criteria linear weighting scheme:

$$\mathcal{S}(u,e) = w_1 \cdot \mathcal{D}(u,e) + w_2 \cdot \mathcal{S}_{\text{skill}}(u,e) + w_3 \cdot \mathcal{R}_{\text{sim}}(u,e) + w_4 \cdot \mathcal{E}_{\text{evid}}(e) + w_5 \cdot \mathcal{A}(u,e) + w_6 \cdot \mathcal{C}(u,e) \quad (1)$$

Subject to the convex optimization constraint:

$$\sum_{i=1}^{6} w_i = 1.0 \quad \text{where} \quad w_i \geq 0 \quad \forall i \in \{1, \ldots, 6\} \quad (2)$$

The individual constituent normalized metrics are defined as follows:

1. **Domain Relevance $\mathcal{D}(u, e) \in [0, 1]$:** Measures taxonomic alignment across the hierarchical academic tree $\mathcal{T}$:
$$\mathcal{D}(u,e) = \frac{2 \cdot \text{depth}(\text{LCA}(d_u, d_e))}{\text{depth}(d_u) + \text{depth}(d_e)} \quad (3)$$
where $\text{LCA}(d_u, d_e)$ denotes the Lowest Common Ancestor of the project's required domain $d_u$ and the expert's primary domain $d_e$ within the knowledge taxonomy.
2. **Skill Compatibility $\mathcal{S}_{\text{skill}}(u, e) \in [0, 1]$:** Evaluates the intersection of required project skills $\mathcal{K}_u$ and the expert's validated skills $\mathcal{K}_e$ using a weighted soft Jaccard index:
$$\mathcal{S}_{\text{skill}}(u,e) = \frac{\sum_{k \in \mathcal{K}_u \cap \mathcal{K}_e} \omega_k}{\sum_{k \in \mathcal{K}_u \cup \mathcal{K}_e} \omega_k} \quad (4)$$
where $\omega_k$ represents the inverse document frequency (IDF) weight of skill $k$ across the global expert corpus.
3. **Semantic Research Interest Similarity $\mathcal{R}_{\text{sim}}(u, e) \in [0, 1]$:** Dense vector cosine similarity between the dense embedding of the student's problem text $\mathbf{v}_u$ and the candidate expert's aggregated research dossier $\mathbf{v}_e$ (composed of bio, publications, and thesis abstracts):
$$\mathcal{R}_{\text{sim}}(u,e) = \max\left(0, \frac{\mathbf{v}_u \cdot \mathbf{v}_e}{\|\mathbf{v}_u\|_2 \|\mathbf{v}_e\|_2}\right) \quad (5)$$
where $\mathbf{v} \in \mathbb{R}^d$ is computed via a fine-tuned dense sentence transformer (e.g., Sentence-BERT [2] or `SciBERT` [7]).
4. **Evidence and Verified Experience $\mathcal{E}_{\text{evid}}(e) \in [0, 1]$:** Quantifies verified credibility rather than self-reported claims:
$$\mathcal{E}_{\text{evid}}(e) = \alpha \cdot V_{\text{badge}}(e) + (1 - \alpha) \cdot \min\left(1.0, \frac{\log(1 + N_{\text{pub}}(e))}{\log(1 + N_{\max})}\right) \quad (6)$$
where $V_{\text{badge}}(e) \in \{0.25, 0.5, 0.75, 1.0\}$ corresponds to the verified tier level, and $N_{\text{pub}}(e)$ is the count of verified peer-reviewed publications.
5. **Availability Compatibility $\mathcal{A}(u, e) \in [0, 1]$:** Evaluates the overlap between the student's expected project schedule $T_u$ (hours/week and duration) and the expert's active availability capacity $T_e$:
$$\mathcal{A}(u,e) = 1.0 - \frac{|T_u - T_e|}{\max(T_u, T_e, 1)} \quad (7)$$

6. **Collaboration Preference Compatibility $\mathcal{C}(u, e) \in [0, 1]$:** Binary indicator or multi-class match evaluating requested collaboration modality (e.g., Virtual, In-Person, Advisory, Co-Author).

### 4.2 Explainable Recommendations (Feature 14)

Trust in AI-driven matching requires complete transparency. For each recommended candidate, UniExpert's Explainable AI (XAI) engine constructs a structured natural language breakdown:

- **Positive Feature Attribution:** Highlighting specific matching skills (e.g., *"Matches 4/4 required skills: PyTorch, YOLOv8, Image Segmentation, Transfer Learning"*).
- **Corroborating Evidence:** Citing specific verified publications (e.g., *"Authored: 'Automated Lesion Detection in Avian Flocks (2024)' in Journal of Poultry Science"*).
- **Identified Gaps:** Explicitly identifying missing or unverified competencies (e.g., *"Note: Candidate has no recorded publications in Veterinary Pharmacology; experience in field data collection is self-reported"*).

### 4.3 Skill Gap Detection Engine (Feature 15)

When constructing multi-member research teams, the platform evaluates collective skill sets using a set-cover algorithm. Given project requirements $\mathcal{K}_u$ and the set of accepted team members $\mathcal{M} = \{e_1, e_2, \ldots, e_m\}$, the missing skill set $\Delta\mathcal{K}$ is computed as:

$$\Delta\mathcal{K} = \mathcal{K}_u \setminus \bigcup_{e \in \mathcal{M}} \mathcal{K}_e \tag{8}$$

If $|\Delta\mathcal{K}| > 0$, the platform immediately invokes targeted sub-searches specifically calibrated to minimize $|\Delta\mathcal{K}|$ with the minimum number of additional recruits.

## 5 System Architecture and Implementation Stack

### 5.1 High-Level Architectural Blueprint

UniExpert adopts a modular, service-oriented architecture ensuring separation of concerns, rapid testing, and independent scaling of heavy ML inference workloads. Figure 2 presents the comprehensive system topology.

### 5.2 Technology Stack Selection and Justification

Table 2 outlines the technological framework selected for the implementation of the UniExpert platform.

## 6 Research Methodology and AI/ML Evaluation Plan

A critical requirement of an academic Software Engineering capstone is rigorous empirical evaluation. UniExpert is designed not merely as a commercial web utility, but as a research-grade artifact testing formal scientific hypotheses.

### 6.1 Primary Research Questions

- **RQ1:** *Does a multi-criteria hybrid recommendation model yield statistically significant higher recommendation accuracy (Precision@5 and NDCG@5) compared to baseline keyword retrieval (BM25) and dense semantic vector similarity alone?*

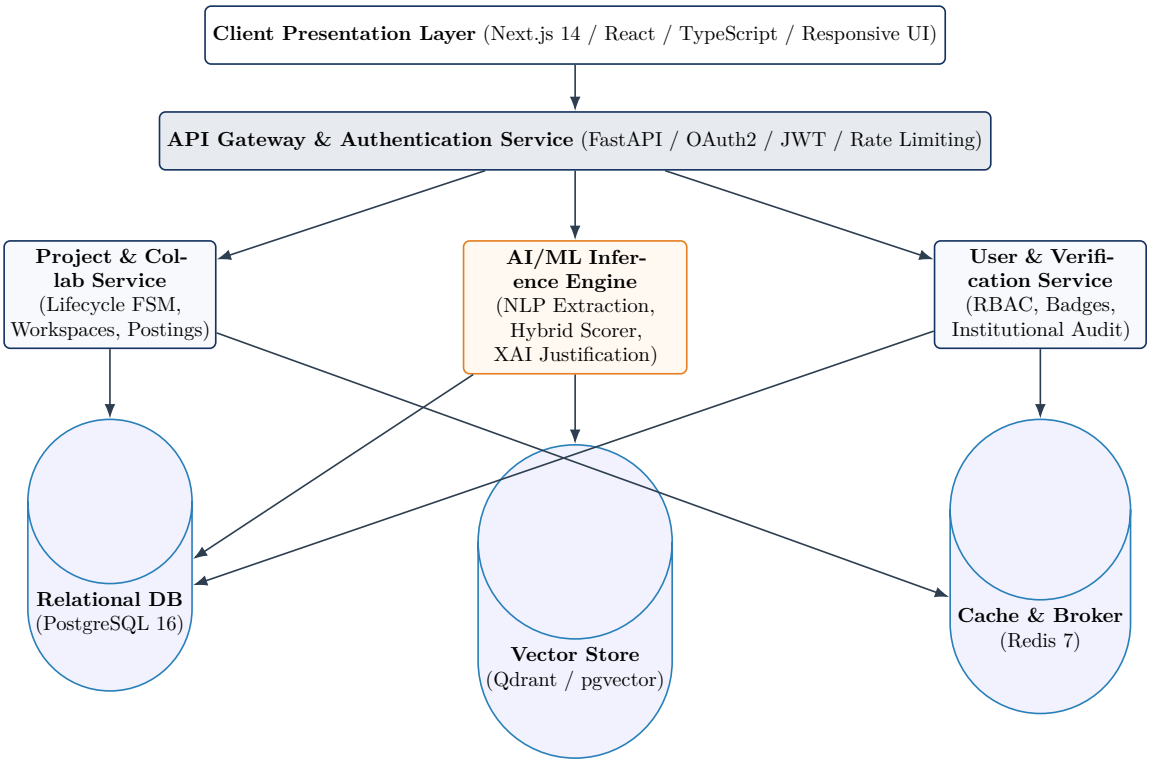



Figure 2: UniExpert High-Level System Architecture

Table 2: Technology Stack Specifications and Architectural Rationale

| Layer | Selected Technology | Engineering Justification |
|---|---|---|
| **Frontend Client** | Next.js 14 / React / TypeScript / Modern CSS | Server-Side Rendering (SSR) for fast loading, strong static typing, reactive component architecture, and modern responsive layouts. |
| **Backend REST API** | FastAPI (Python 3.11) | Native asynchronous execution, automated OpenAPI documentation, high throughput, and seamless integration with Python ML pipelines. |
| **AI & NLP Core** | HuggingFace Transformers, PyTorch, SciBERT | State-of-the-art transformer architectures for domain classification, dense sentence embeddings, and token-level skill extraction. |
| **Dense Vector Store** | Qdrant / `pgvector` | Ultra-low latency Approximate Nearest Neighbor (ANN) vector indexing via HNSW graphs; scales to hundreds of thousands of expert vectors. |
| **Primary Database** | PostgreSQL 16 | ACID compliance, rich JSONB support for dynamic profile portfolios, and robust relational integrity for user authentication and request FSMs. |
| **Caching & Queues** | Redis 7 | Session state caching, rate limiting, and asynchronous message broker for background embedding indexing jobs. |

- **RQ2:** *To what degree does automated natural language skill extraction align with expert human-annotated requirements across interdisciplinary capstone problem statements?*
- **RQ3:** *Do explainable recommendation justifications increase student trust and collaboration request conversion rates compared to unadorned similarity scores?*

### 6.2 Experimental Baseline Comparisons

To evaluate the recommendation engine objectively, we implement and benchmark three competing retrieval pipelines under identical corpus conditions:

1. **Baseline A (Lexical Matching):** Classical BM25 keyword search over tokenized expert profile resumes and publication titles.
2. **Baseline B (Dense Semantic Retrieval):** Pure bi-encoder sentence embedding cosine similarity ($\mathcal{R}_{\text{sim}}$) without skill weighting or evidence filtering.
3. **Proposed Method (UniExpert Hybrid):** Multi-criteria weighted ranking engine incorporating domain taxonomy, skill IDF Jaccard, semantic similarity, verified evidence tiers, and schedule compatibility (Eq. 1).

### 6.3 Evaluation Metrics

The quantitative performance of the retrieval pipelines will be benchmarked using the standard Information Retrieval (IR) metrics defined below:

Table 3: Formal Evaluation Metrics for Recommendation Performance

| Metric | Mathematical Definition | Evaluation Objective |
|---|---|---|
| **Precision@$K$** | $\text{P@}K = \frac{\lvert\text{Relevant Experts in Top-}K\rvert}{K}$ | Measures the proportion of relevant experts appearing in the top-$K$ recommendations ($K \in \{5, 10\}$). |
| **Recall@$K$** | $\text{R@}K = \frac{\lvert\text{Relevant Experts in Top-}K\rvert}{\lvert\text{All Known Relevant Experts}\rvert}$ | Measures the engine's ability to retrieve all suitable candidates from the dataset. |
| **NDCG@$K$** | $\text{NDCG@}K = \frac{\text{DCG@}K}{\text{IDCG@}K}$ where $\text{DCG@}K = \sum_{i=1}^{K} \frac{2^{rel_i}-1}{\log_2(i+1)}$ | Evaluates ranking quality, heavily penalizing relevant experts placed low in the result list. |
| **Skill Extraction F1** | $\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision}+\text{Recall}}$ | Evaluates the NLP token extraction accuracy against human gold-standard annotations. |
| **System Latency** | $p_{50}, p_{95}, p_{99}$ response times (ms) | Validates real-time responsiveness and engineering throughput of the vector index. |

### 6.4 Dataset Construction and Human Relevance Judgment

To avoid synthetic bias, an evaluation dataset comprising 100 authentic, anonymized capstone problem descriptions spanning five faculties (Engineering, Agriculture, Business, Health Sciences, and Humanities) will be constructed. A pool of 300 consenting expert profiles (professors, researchers, alumni) will be seeded. A panel of qualified academic evaluators (senior faculty members and researchers) will perform blind graded relevance assessments on a 3-point Likert scale (0: Irrelevant, 1: Partially Relevant, 2: Highly Relevant). Inter-rater agreement will be validated using Cohen's Kappa ($\kappa \geq 0.70$).

## 7 Trust, Safety, Privacy, and Algorithmic Fairness

## 7.1 Anti-Bias and Fair Matching Principles (Feature 25)

A recognized hazard in academic recommender systems is the *Matthew Effect* ("rich get richer"), wherein senior, highly cited scholars receive all recommendation traffic while junior faculty and early-career researchers remain undiscovered. UniExpert incorporates explicit algorithmic fairness safeguards:

- **Popularity De-biasing:** The scoring function avoids raw profile view counts or global popularity as an input metric.
- **Cold-Start Exploration Boost:** A randomized contextual $\epsilon$-greedy exploration mechanism allocates 10% of top recommendation slots to newly registered, fully verified experts possessing high skill overlap.
- **Protected Attributes Blindness:** Sensitive personal attributes (gender, age, ethnicity, national origin) are strictly excluded from all vector indexing and ranking formulas.

## 7.2 Privacy Controls and Data Governance (Feature 23)

All platform interactions adhere strictly to academic data protection standards. Students can designate project descriptions as Private or Protected. Direct contact details (telephone numbers, personal emails) are masked behind the platform's workspace abstraction until explicit collaboration acceptance is confirmed by both parties.

# 8 Implementation Scope and MVP Prioritization

To guarantee successful project delivery within the Fall 2026 semester timeframe (October – December 2026, 12 weeks), features are strictly triaged using the MoSCoW prioritization model, as detailed in Table 4.

Table 4: MoSCoW Feature Scope Prioritization for Fall 2026 (October – December 2026)

| Classification | Feature Scope & Description | Delivery Target |
|---|---|---|
| **Must-Have** (Core MVP) | • Student and expert registration with role-based auth.<br>• Rich student and expert profiles with skill taxonomy.<br>• Multi-tier expert verification workflow (email & documents).<br>• Natural language project submission & NLP skill extractor.<br>• Semantic dense vector search and hybrid multi-criteria ranker.<br>• Explainable natural language recommendation justifications.<br>• Collaboration request management (FSM) & basic messaging.<br>• Administrative supervision dashboard & empirical evaluation. | **October – November 2026** (Weeks 1–8) |
| **Should-Have** (Extended Scope) | • Multidisciplinary problem decomposition & team builder.<br>• Automated skill gap detection for project teams.<br>• Side-by-side expert comparison matrix.<br>• Automated Google Scholar / ORCID public metadata sync.<br>• Structured post-collaboration ratings and feedback. | **December 2026** (Weeks 9–12) |
| **Could-Have** (Future Work) | • Integrated real-time WebRTC video conference rooms.<br>• Institutional cross-campus analytics federation.<br>• Paid consultation escrow and honorarium management.<br>• Native mobile applications (iOS/Android). | Post-Capstone / Future Research |

# 9 Semester Development Plan and Milestones (Fall 2026)

The capstone project spans a disciplined 12-week Fall 2026 semester execution schedule (October – December 2026), structured across three distinct monthly milestones, as synthesized in Table 5.

Table 5: Fall 2026 (October – December 2026) Capstone Development Timeline and Deliverables

| Phase | Duration | Key Engineering Deliverables & Research Activities | Milestone |
|---|---|---|---|
| **Month 1** (October 2026) | Weeks 1–4 | • Formalize Software Requirements Specification (SRS) and architecture.<br>• Literature review of expert finding and recommender systems.<br>• Stakeholder interviews with faculty advisors and student researchers.<br>• PostgreSQL relational schema, Redis cache design, and API contracts.<br>• Git repository governance, automated linting, and CI/CD pipelines. | **M1: System Blueprint** (Requirements & Architecture) |
| **Month 2** (November 2026) | Weeks 5–8 | • Implement OAuth2/JWT authentication and RBAC security services.<br>• Student and expert profile schemas, portfolio UI, and verification.<br>• Natural language project parsing and NLP skill extraction pipeline.<br>• Qdrant vector database integration with dense sentence embeddings.<br>• Hybrid multi-criteria ranking model (Eq. 1) and XAI explanations.<br>• Collaboration request state machine (FSM) and workspace channels. | **M2: Core Platform & AI Matchmaking Engine** |
| **Month 3** (December 2026) | Weeks 9–12 | • Multidisciplinary team builder and skill gap optimizer (Eq. 8).<br>• Empirical IR comparative evaluation (BM25 vs. Vector vs. Hybrid).<br>• System integration, end-to-end security audits, and UAT with users.<br>• Comprehensive capstone thesis report, documentation, and final defense. | **M3: Empirical Evaluation & Final Defense** |

## 10 Risk Assessment and Mitigation Strategies

A rigorous engineering proposal must preemptively identify technical and organizational hazards. Table 6 formalizes the risk matrix and corresponding mitigation protocols.

Table 6: Comprehensive Project Risk Matrix and Mitigation Strategies

| Identified Risk | Likelihood | Impact | Engineered Mitigation Protocol |
|---|---|---|---|
| **Data Cold-Start** (Scarce initial profiles) | High | High | Pre-seed database with public faculty profiles from consenting departments; implement automated ORCID/Google Scholar import utilities. |
| **NLP Extraction Hallucination** | Med | Med | Combine regex-based taxonomy dictionary matching with zero-shot transformer classifiers; provide interactive UI allowing students to verify extracted skills. |
| **Vector Search Latency** | Low | High | Pre-compute embeddings asynchronously; utilize HNSW indexing in Qdrant; cache top domain search clusters in Redis. |
| **Credential Falsification** | Med | High | Enforce institutional email verification (`.edu`); display distinct, unforgeable badge indicators separating verified records from claims. |
| **Faculty Availability Fatigue** | High | Med | Implement monthly request quotas per expert; provide automated calendar blackout toggles; enable one-click polite declination templates. |

## 11 Conclusion and Expected Capstone Outcomes

The **UniExpert** platform addresses an urgent, unfulfilled need in higher education: breaking down academic silos to connect university researchers and students through transparent, AI-driven expert discovery. By departing from rigid keyword indexes and pioneering a problem-driven, multi-criteria recommendation architecture, UniExpert empowers students to find the exact mentorship necessary to elevate their capstone projects from standard coursework into high-impact academic contributions.

Upon successful completion of this Fall 2026 capstone project (October – December 2026), the expected academic and engineering deliverables comprise:

1. A fully functional, production-ready, accessible web platform conforming to modern software engineering standards.
2. An empirical research dataset and benchmarking suite establishing quantitative improvements in recommendation quality over classical IR baselines.
3. A comprehensive capstone thesis report and a submission-ready research paper targeted at academic software engineering and educational technology venues.
4. A scalable foundation for institutional adoption across university departments.

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